

- **“Active learning accelerates electrolyte solvent screening for anode-free lithium metal batteries”** – Peiyuan Ma et al., Nat. Commun. **16**, 8396 (2025), DOI: 10.1038/s41467-025-63303-7. This study uses a **Bayesian active - learning** workflow to discover new liquid electrolyte solvents. Starting from only 58 measured electrolytes, the model iteratively screened a virtual library of 1 million candidates. After 7 rounds of “suggest - test - update” cycles (~10 experiments per round), the algorithm identified four novel electrolyte solvents whose anode - free Li|LiFePO₄ cell performance rivaled state-of-the-art systems ¹. Key to this approach was integrating experimental feedback (cell capacity retention) into a sequential design algorithm (Bayesian model averaging) to efficiently explore the sparse, high-dimensional chemical space ¹.
- **“Chemical foundation model-guided design of high ionic conductivity electrolyte formulations”** – Murtaza Zohair et al., NPJ Comput. Mater. **11**, 283 (2025), DOI: 10.1038/s41524-025-01774-4. Here a large pretrained **chemical language (foundation) model** was fine-tuned on 13,666 experimental conductivity data points for Li - ion battery electrolytes ². The model was then used in a **generative-screening** framework to propose new high - conductivity formulations. Seven novel electrolyte mixtures (combining carbonate solvents with LiFSI or LiDFOB salt) were discovered, with ionic conductivities improved by ~82% (LiFSI - based) and ~172% (LiDFOB - based) over the baseline ². This work demonstrates how a generalizable AI model can navigate vast formulation space and suggests specific solvent mixtures for experimental validation ².
- **“Machine-Learning-Guided Screening of Advantageous Solvents for Solid Polymer Electrolytes in Lithium Metal Batteries”** – Jiadong Shen et al., Nano Lett. **25**, 7801–7809 (2025), DOI: 10.1021/acs.nanolett.5c00797. This work combines high-throughput DFT and ML to screen ~10,000 candidate solvent molecules (C/H/O/N/F constituents) for use as residual solvents in PVDF - HFP polymer electrolytes ³. The authors trained ensemble ML models (XGBoost, SISO, CGCNN) on molecular descriptors (HOMO/LUMO, polarity, donor number, etc.) to predict which solvents yield high ionic conductivity and stable SEI. Two top candidates emerged: 2,2,2 - trifluoro - N,N - dimethylacetamide (TFDMA) and N-methoxy-N-methyl-2,2,2-trifluoroacetamide (TFOMA). Experimentally adding trace TFOMA to PVDF-HFP gave a wide electrochemical window (>4.5V), high conductivity ($5.5 \times 10^{-4} \text{ S cm}^{-1}$ at 30°C) and large Li⁺ transference number (0.78) ³. LiFePO₄||Li and NCM91||Li full cells with PVDF-HFP@TFOMA retained ~98–99% capacity over hundreds of cycles, far outperforming cells with conventional casting solvents (DMF, NMP, DMSO) ³ ⁴. This integrated DFT+ML→experiment pipeline establishes solvent design rules and validates a superior new polymer electrolyte solvent ³ ⁴.
- **“Machine Learning–Driven Mass Discovery and High-Throughput Screening of Fluoroether-Based Electrolytes for High-Stability Lithium Metal Batteries”** – Qingqing Jia et al., Angew. Chem. Int. Ed. **64**, e202424493 (2025), DOI: 10.1002/anie.202424493. In this study, a hybrid ML framework (voting ensemble + graph convolutional neural net) was built to evaluate thousands of candidate fluoroether solvents for oxidative stability, salt solvating power, and synthetic feasibility ⁵ ⁶. A virtual library of 5,576 fluoroether/salt combinations was screened. Two promising fluorinated ether molecules (1,1,1,3,3,3-hexafluoro-2-(2-methoxyethoxy)propane and 7,7,8,8-tetrafluoro-3,12-dimethoxy-2,5,10,13-tetraoxatetradecane) were selected, synthesized, and tested. These new solvents showed excellent high-voltage stability and solvation; Li|NMC811 cells using them delivered ~99.5% Coulombic efficiency and stable cycling ⁵ ⁶. This work illustrates a “screen-first” AI workflow: predictive models identify novel fluoroether solvents, then targeted experiments confirm their superior electrolyte performance ⁵ ⁶.

Each of these papers exemplifies a **closed-loop discovery workflow**: machine learning or generative models suggest new electrolyte solvent (or formulations) candidates, which are then synthesized and validated electrochemically. All incorporate iterative ML screening of large chemical spaces with experimental confirmation of at least the top candidates ¹ ⁵ .

Sources: Recent publications (2023–2025) from Nature Commun., npj Comp. Mater., Nano Lett., Angew. Chem. Int. Ed. etc., detailing AI/ML-driven electrolyte solvent design and experimental validation ¹ ² ³ ⁵ .

¹ Active learning accelerates electrolyte solvent screening for anode-free lithium metal batteries | Nature Communications

https://www.nature.com/articles/s41467-025-63303-7?error=cookies_not_supported&code=dd0616ff-3128-455f-9ee1-aaf4ee12d4f8

² Chemical foundation model-guided design of high ionic conductivity electrolyte formulations | npj Computational Materials

https://www.nature.com/articles/s41524-025-01774-4?error=cookies_not_supported&code=fdb7ee48-b9aa-40c3-aa54-5017196efecb

³ ⁴ Machine-Learning-Guided Screening of Advantageous Solvents for Solid Polymer Electrolytes in Lithium Metal Batteries - PMC

<https://pmc.ncbi.nlm.nih.gov/articles/PMC12082693/>

⁵ ⁶ Machine Learning-Driven Mass Discovery and High-Throughput Screening of Fluoroether-Based Electrolytes for High-Stability Lithium Metal Batteries - PubMed

<https://pubmed.ncbi.nlm.nih.gov/39930920/>